

AQI measurement system using UAVs

-for the detection of gases and pollution closely

Gas sensing using Arduino:

While understanding having so many gases in our environment like PM2.5, PM10, NO₂, O₂, CO₂, CO, Methane and natural gasses we tried to restore from natural we using DFRobot for nitrogen oxide and carbon monoxide . NCD.io for methane the arduino makes a graph output by all datasets from the sensors.

Uav for pollution monitoring

A UAV carrying pollution-monitoring sensors can fly (either autonomously or via human control) to multiple waypoints while repetitively gathering pollution and location data as graph.

This data can be sent back to the human controller, who can then use the data to predict potential pollution sources and monitor pollution trends.

Which helps us to prevent pollution and maintain its aqi index

Challenges with development process:

- Incompatibility of hardware which forced to mount on complicated intrinsic designs
- Software tuning to be used from scratch .
- Drone reengineering while components adjustment with angle and new mountings on drone

Limitations for UAV systems
The current UAV design faces several limitations that affect data quality and operational performance. The low-cost GPS module (M8) exhibits an average positional error of approximately 5 meters, which is further influenced by electromagnetic interference generated by high-current

motors. Similarly, the metal-oxide semiconductor air-quality sensors are sensitive to temperature and humidity variations, resulting in fluctuating readings that provide only relative measurements rather than precise gas concentrations. The compact drone platform also limits battery capacity, restricting flight endurance to approximately 10 minutes and reducing the area that can be surveyed in a single mission. Additionally, the onboard flight controller and radio communication system have limited processing and transmission capabilities, preventing real-time transfer of sensor data. As a result, only essential telemetry such as GPS position, battery voltage, motor RPM, and internal temperature can be transmitted during flight, while environmental sensor data must be stored onboard and analyzed post-flight. Future improvements include higher-accuracy sensors, enhanced communication systems, and increased battery capacity to improve reliability, data accuracy, and mission coverage.

And designing this aqi model for sensors to measure temperature , humidity , carbon monoxide , nitrogen oxide , hydrogen, and methane content in patch area researching for it

WSN Wireless Sensor Network
COMPRISED-WITH
MULTIPLE NODES

SENSOR CLASSIFICATION

For gas detection, there is a wide variety of devices based on different kinds of materials and the principles of operation. Gas sensors are classified into many kind of sensor types; various approaches can be used. For example, optical sensors, MQ4 sensor, MQ8 sensor, MQ9 sensor, MQ135 sensor, and DHT22 sensors. In this project, the sensors will be chosen based on the desired range and constraints of the design such as accuracy, heating time and response time. For instance, an electrochemical sensor is used for gases, dust is an optical sensor, temperature and humidity sensor is an electrical sensor

OZONE

Sampling at the location will provide the best indication for efficacy of the O₃ catalyst and of the worst case for occupant exposure, in that supply air is typically diluted with recirculated air

CO

Air supply ducts leading to each of the cabin ventilation zones. Electrochemical devices can be operated in the active mode, in which a small air pump draws a continuous sample from the air supply ducts to the sensor is directly in the air stream to be sampled and CO enters the sensor by diffusion through a semipermeable membrane.

CO₂

Exhaust ducts from the ventilation zones of aircraft to monitor effectiveness of fresh air ventilation to each zone. Effective ventilation depends on the volume of fresh air delivered to the space and the extent to which the incoming air is mixed with the resident air by turbulence and natural convection.

Humidity

Rel-humidity sample the air leaving from the cabin and cockpit because the outside air at cruise altitude contains essentially no moisture, the moisture

content of the cabin air will provide an indication of the balance between the indoor sources of moisture.

PM sensor:

Particulate Matter (PM) Sensors monitor the function of the Diesel Particulate Filter (DPF) to detect excessive PM emissions. This sensor is used to monitor the function of the diesel particle filter (DPF) and helps reduce particulate emissions by up to 99 %. We are using the frequency of 868 Hz.

ESC (Electronics Speed Controller) It is connected to the motor for controlling the speed. Esc will be connected to the flight controller. An electronic speed control (ESC) is an electronic circuit that controls and regulates the speed of an electric motor. It may also provide reversing of the motor and dynamic braking. Miniature electronic speed controls are used in electrically powered radio controlled models. Full-size electric vehicles also have systems to control the speed of their drive motors.

RECEIVING-SENSOR SIGNALS

- Wireless communication between the UAV and Ground Control Station (GCS) is established using radio antennas configured through dedicated software. Each antenna is assigned a unique network address to enable reliable bidirectional data transmission between the aerial and ground systems.

- The communication modules are interfaced with NodeMCU controllers through quick-connect shield modules. The NodeMCU onboard the UAV (Node 1) is configured in **Router Mode**, enabling it to collect sensor data and forward telemetry information during flight.

- The Ground Control Station node is configured in **Coordinator Mode**,

allowing it to receive, process, and monitor data transmitted from the UAV in real time.

- Sensor information, including GPS coordinates, battery voltage, motor RPM, and environmental measurements, is acquired by the onboard controller and transmitted through the wireless communication link for monitoring and analysis.
- This communication architecture provides reliable telemetry exchange between the UAV and the ground station, supporting remote monitoring, navigation, and mission management operations.

Reference

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AQI Measurement books